MARKETIX

Status: Ready for Submission
Topic: Practical App Guide

Practical User Guide & Core Concept (Easy Language)

1. What is MARKETIX? (In Plain Words)

MARKETIX is an intelligent software app created for retail stores and businesses to stop wasting money on useless marketing.

Instead of blindly sending email blasts, text messages, or paper flyers to *all* customers, MARKETIX analyzes your customer data to tell you:
• **Who your best customers are** and what products they actually purchase.
• **Which group they belong to** (e.g. affluent high spenders vs bargain deal hunters).
• **Whether a specific customer will actually buy** from your next campaign or ignore it (before spending any money).

2. The 4 Main Practical Uses of MARKETIX

Use Case	What It Does	Business Benefit
1. Spending & Channel Analytics	Analyzes sales across 6 product types (Wines, Meat, Fruits, etc.) and 3 channels (Web, Catalog, Physical Stores).	Shows which channels drive the most revenue so companies know where to invest.
2. Automatic Customer Segmentation	Uses Machine Learning (K-Means) to divide 2,240 customers into 4 clear customer personas.	Stops one-size-fits-all marketing. Tailors products to each customer's budget and lifestyle.
3. Campaign Response Predictor	Allows staff to enter customer details and get an instant AI probability score (0% to 100%) of buying.	Saves up to 40% in campaign costs by filtering out people who will never buy.
4. 1-Click PDF Executive Reports	Compiles all metrics, charts, and recommendations into a formatted PDF file at the click of a button.	Saves hours of manual Excel reporting for team leads and executives.

3. How to Use the App (Step-by-Step Practical Guide)

Here is how anyone uses MARKETIX from start to finish with concrete examples:

Step 1: Open the Application

- Run the app in your browser at http://localhost:8501.
  - The landing page immediately shows 6 primary KPI cards: **2,240 Total Customers**, **5 Historical Campaigns**, **334 Positive Responses (14.91% rate)**, **$52,243 Average Income**, and **$606 Average Spend**.

Step 2: Check Overall Performance (Click 'Dashboard')

- Click **Dashboard** in the sidebar to view revenue distribution.
  - *Practical Discovery:* You will observe that **Wine ($304 avg)** and **Meat ($167 avg)** make up over 75% of total sales, and customers make twice as many purchases in **Physical Stores** compared to online.
  - *Decision:* Keep premium wine and meat prominently displayed in physical store entrances.

Step 3: Group Your Customers (Click 'Customer Segmentation')

- Click **Customer Segmentation** on the sidebar, keep clusters at 4, and click **Run Clustering**.
  - The AI groups all 2,240 customers into 4 distinct groups:

Persona Group	Income / Profile	What They Buy	Action to Take
Affluent Connoisseurs	Income: $78,000+ No young children	Wines ($600+) & Meats ($400+)	Send invitations to VIP premium wine tastings (no discount needed).
Family Value Seekers	Income: $35k - $55k 1 to 3 children	Groceries, sweets, mid-tier items	Send family-pack grocery bundles and school season promotions.
Digital Natives	Income: $40k - $65k Younger demographic	Frequent web & catalogue purchases	Target with online app flash sales and fast delivery perks.
Budget Deal Shoppers	Income: Below $35,000	Purchases only with promo deals	Send clearance deal notifications to move unsold inventory.

Step 4: Test a Customer Before Spending Money (Click 'Response Predictor')

This is the core predictive engine. Instead of guessing, staff can test customer profiles:

Scenario	Inputs Entered	AI Prediction	Action Taken
Customer A (Active Buyer)	• Income: **$82,000** • Recency: **12 days** • Past accepted: **2** • Web visits/mo: **3**	**87.2% Likelihood** HIGH PROPENSITY	**SEND OFFER** High probability of conversion and strong profit.

Customer B (Dormant Buyer)	• Income: **$24,000** • Recency: **85 days** • Past accepted: **0** • Web visits/mo: **8**	**4.1% Likelihood** LOW PROPENSITY	**SUPPRESS OFFER** Save print/postage cost. Do not spend budget.

Step 5: Export Professional Report (Click 'Reports')

- Click **Reports** in the sidebar.
  - Click **Generate PDF Report**.
  - Download the file to present to managers, executives, or examiners.

4. Real-Life Example: How MARKETIX Saves $35,000 on a Campaign

(You can say this exact 30-second explanation to your teacher):

*"Teacher, imagine a store wants to mail 10,000 printed promotional catalogs. Printing and mailing each catalog costs $5, so the campaign costs **$50,000**.*

*Without MARKETIX, they mail to everyone. Usually only 5% respond, meaning **$47,500 is completely wasted** on 9,500 people who threw the catalog straight in the trash.*

With MARKETIX, we run our customer list through our Machine Learning model first. The model identifies the 7,000 people who have almost zero chance of responding and filters them out. We only print and mail catalogs to the 3,000 high-probability customers.

***The Result:** Marketing costs drop from **$50,000 down to $15,000 (saving $35,000)**, while still capturing almost all the sales!"*

Key Takeaway: MARKETIX transforms marketing from blind guessing into data-driven, profitable decision-making.